**Name of the Student:** _____ **Roll no:** _____

**CLASS: - T.E (CSE)** **Division:** **Course: - PCC305CSE- Machine Learning Lab**

**Experiment No. 06**

**** Binary Classification Using Logistic Regression****

Marks: /10

**Date of Performance:** / /2026 **Sign with Date:**

**Aim:** Binary Classification Using Logistic Regression.

**Lab Objective**: To build and evaluate a Logistic Regression model for binary classification on a real-world open-access dataset, and to assess its diagnostic performance using a Confusion Matrix, Precision, Recall, and F1-Score..

## Problem Statement:

Illustrate the performance of a binary classification model using logistic regression on a suitable open-access dataset through a confusion matrix, precision, recall, and F1-score.

## 1. Theory:

Logistic Regression & The Sigmoid Function

Unlike linear regression, Logistic Regression models the probability P(Y=1|X) that an observation belongs to a positive class using the Sigmoid (Logistic) Function:

$$P(Y=1|X)=\sigma(z)=1+e-z1$$

Where the logit z is a linear combination of input features:

$$z=\beta_0 +\beta_1 X_1 +\beta_2 X_2 +\cdots+\beta_p X_p$$

- If P(Y=1|X)≥0.5, the prediction y^ =1 (Positive).
- If P(Y=1|X) <0.5, the prediction y^ =0 (Negative)

## 2. The Confusion Matrix

A tabular layout that cross-tabulates actual vs. predicted labels:

| | **Predicted Negative ($y$^=0)** | **Predicted Positive ($y$^=1)** |
|---|---|---|
| **Actual Negative ($y$=0)** | **True Negative (TN)** | **False Positive (FP)** *(Type I Error)* |
| **Actual Positive ($y$=1)** | **False Negative (FN)** *(Type II Error)* | **True Positive (TP)** |

## 3. Classification Evaluation Metrics

- **Precision (Positive Predictive Value):** Of all instances predicted as positive, how many were truly positive?

$$\textbf{Precision=TP+FPTP}$$

- **Recall (Sensitivity / True Positive Rate):** Of all actual positive cases, how many did the model correctly catch?

$$\textbf{Recall=TP+FNTP}$$

- **F1-Score (Harmonic Mean):** Combines Precision and Recall into a single metric, especially useful when class distributions are imbalanced:

$$\textbf{F1-Score=2×Precision+RecallPrecision×Recall=2TP+FP+FN2TP}$$

- **Accuracy:** Overall proportion of correct predictions:

$$\textbf{Accuracy=TP+TN+FP+FNTP+TN}$$

## 4. Dataset Selection: Breast Cancer Wisconsin (Diagnostic)

- **Source:** UCI Machine Learning Repository (available directly via sklearn.datasets.load_breast_cancer)
- **Samples ($N$):** 569 instances
- **Features ($p$):** 30 continuous cell nucleus measurements (e.g., radius, texture, perimeter, area, smoothness, concavity)
- **Classes:**
    - 0: Malignant (Cancerous)

    - 1: Benign (Non-cancerous)

## 5. Breakdown of the Confusion Matrix:

- **True Negatives (TN):** 41 Malignant cases correctly flagged as Malignant.
- **False Positives (FP):** 1 Malignant case incorrectly classified as Benign (Type I error).
- **False Negatives (FN):** 1 Benign case incorrectly classified as Malignant (Type II error).
- **True Positives (TP):** 71 Benign cases correctly classified as Benign.

## Questions:

1. Why use the Sigmoid function instead of a standard linear regression line for classification?
2. In medical diagnosis, which metric is generally prioritized: Precision or Recall? Why?
3. Why is the harmonic mean used for the F1-Score instead of the standard arithmetic mean?
4. What is the impact of changing the classification probability threshold from 0.5 to 0.3?
5. Why must features be standardized before training Logistic Regression?

| Attendance and Timely Submission (10) | Execution (10) | Observations and Records (10) | Journal Documentation (10) | Oral (10) |
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**Signature of Subject Teacher**